

Write-Behind Worker with Crash Simulation

 systemdesignschool.io/fundamentals/cache-lab-write



Lab: The Write Drill

The previous article explained why caches and databases can disagree. This lab implements a write-behind worker from scratch so you can see write coalescing and the data loss risk in working code.

Exercise 1: Write-Behind Worker

Implement a write-behind cache that buffers writes in memory and flushes to the "database" in batches. The flush happens when the buffer reaches a threshold or when you trigger it manually.

Write-Behind: Buffered Flush

Notice how all 5 writes complete instantly (sub-millisecond each). The flush happens once and writes all pending entries to the database. During the window between writes and flush, the database is stale—but reads from the cache always return the latest value.

Exercise 2: Write Coalescing

When the same key is written multiple times before a flush, only the final value needs to reach the database. This reduces database writes dramatically for frequently-updated keys.

Write Coalescing: Same Key, One DB Write

The buffer uses a dictionary instead of a list. Each new write to the same key overwrites the previous value. On flush, only one database write per key is needed regardless of how many times it was updated.

Exercise 3: Crash Simulation

The risk of write-behind is data loss. If the process crashes before flushing, buffered writes are gone. This exercise simulates that failure.

Crash Before Flush: Data Loss

The crash wipes the in-memory buffer. All writes between the last flush and the crash are permanently lost. The mitigation—flushing more frequently—reduces the window of vulnerability but never eliminates it entirely. This is the fundamental tradeoff of write-behind: speed in exchange for durability risk.

For data where loss is unacceptable, use write-through (synchronous) or write-around with immediate database writes. Write-behind is appropriate only when the application can tolerate losing the most recent unflushed writes—session data, analytics counters, or recommendation scores where approximate values are acceptable.